PRODUCT REQUIREMENT DOCUMENT (PRD)

UMHackathon 2026

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Project Name: StockWise — AI Inventory Optimizer

Version: 1.0

1. Project Overview

1.1 Problem Statement

Malaysian SME traders in the import and distribution sector face increasingly complex inventory decision-making challenges driven by fluctuating exchange rates, global supply chain disruptions, and volatile demand cycles. Currently, inventory management systems rely almost exclusively on rule-based threshold mechanisms such as triggering reorder alerts only when stock falls below a predefined limit. While effective in basic scenarios, these systems operate in a vacuum, completely failing to incorporate critical external signals.

In reality, SMEs must continuously respond to dynamic, unstructured information. A sudden tariff announcement, a factory shutdown due to fire, or a sharp MYR depreciation against the USD can dramatically alter procurement costs and product availability overnight. Existing tools lack the capability to interpret these unstructured signals, let alone integrate them with structured operational data like current stock levels or historical sales.

As a result, SMEs operate reactively rather than proactively. Unexpected events routinely invalidate existing inventory plans, leading to overstocking of declining products, understocking of fast-moving items, and inefficient capital allocation. This highlights a critical gap in the market: the absence of an intelligent decision-support system capable of synthesizing both structured data and unstructured market intelligence to generate timely, context-aware inventory strategies.

1.2 Target Domain

The target domain for StockWise is AI-powered supply chain optimization, specifically designed for **Malaysian SME import and trading businesses**. These businesses typically operate with limited IT budgets, small-to-medium warehouse footprints, and rely on importing goods from major manufacturing hubs such as China, the European Union, and the United States. The domain encompasses inventory planning, procurement decision-making, multi-currency cost

management, and supply chain risk mitigation. By focusing on this niche, StockWise addresses a tangible pain point that directly impacts the profitability and competitiveness of Malaysian SMEs in the global trade ecosystem.

1.3 Proposed Solution Summary

StockWise is an AI-powered decision intelligence platform designed to help Malaysian SMEs optimize inventory replenishment under dynamic market conditions. The system integrates structured operational data—such as inventory levels, sales velocity, exchange rates, and supplier lead times—with unstructured external signals, including curated news on supply chain disruptions, trade policy changes, and commodity trends.

At the core of StockWise is Z.AI's GLM, which functions as the primary reasoning engine. Instead of relying on static threshold-based alerts, the system analyzes cross-domain data to generate context-aware inventory recommendations. For example, it can recommend increasing order quantities in anticipation of rising demand or accelerating procurement when currency depreciation is expected to increase import costs.

The platform focuses on actionable decision support by producing clear recommendations, supported by explanations and key influencing factors. Each output includes a suggested action (e.g., reorder quantity adjustment), a justification based on both structured and unstructured inputs, and an estimated impact such as cost savings, reduced stockout risk, or improved inventory turnover.

By transforming fragmented data into explainable, AI-driven strategies, StockWise enables SMEs to shift from reactive inventory management to proactive, data-informed decision-making, ultimately improving cost efficiency and operational resilience.

2. Background & Business Objective

2.1 Background of the Problem

Malaysian SMEs that depend on importing goods from international suppliers operate in an environment characterized by volatile exchange rates, frequent supply chain disruptions, and rapidly shifting demand patterns. The Malaysian Ringgit's fluctuation against major currencies such as the US Dollar, Euro, and Chinese Yuan directly impacts procurement costs, often by margins of 5–15% within a single quarter. Simultaneously, global events — from pandemic-induced factory shutdowns to geopolitical tensions affecting shipping lanes — create sudden and unpredictable supply constraints.

The fundamental challenge of this domain lies in the timing and financial mechanics of international trade. SMEs must commit capital weeks or months in advance based on supplier lead times, creating a dangerous gap between placing a purchase order and the actual arrival of goods. During this lag, currency values can shift, turning a projected profit margin into a loss. Furthermore, warehouse space in Malaysia is expensive; holding excess buffer stock ties up vital cash flow, while holding too little risks stockouts during peak retail seasons. Decision-makers are forced to manually balance these competing financial variables—Foreign Exchange exposure, lead time risks, holding costs, and stockout penalties—often relying on fragmented information and intuition rather than a unified view of their economic reality.

2.2 Importance of Solving This Issue

Research in supply chain analytics consistently demonstrates that early action on supply risks can save organizations between 15% and 30% on annual procurement costs. For Malaysian SMEs operating on thin margins — typically 5–12% — these savings can represent the difference between profitability and loss. By leveraging Z.AI's reasoning capabilities to cross-reference real-time market news with inventory and sales data, StockWise enables proactive decision intelligence that anticipates problems before they materialize, rather than scrambling to respond after the fact.

Furthermore, the ability to reason across multiple data dimensions simultaneously — considering not just stock levels but also currency trends, supplier reliability, seasonal demand patterns, and emerging market risks — provides a competitive advantage that was previously available only to large enterprises with dedicated supply chain analytics teams. StockWise democratizes this capability, making enterprise-level strategic planning accessible to SMEs through an intuitive, affordable web-based platform.

2.3 Strategic Fit

StockWise aligns with Malaysia's national digital transformation agenda, particularly the Malaysia Digital Economy Blueprint (MyDIGITAL), which emphasizes the adoption of AI and data-driven technologies among SMEs to improve productivity and global competitiveness. Supply chain optimization represents a high-impact domain within this initiative, as many Malaysian SMEs depend heavily on international trade and are directly affected by global market fluctuations.

Beyond policy alignment, StockWise demonstrates a practical and applied use of AI for economic empowerment and decision intelligence. Unlike typical chatbot-based implementations, the system utilizes Z.AI's GLM as a core reasoning engine to support complex decision-making. It integrates structured data such as inventory levels and exchange rates with unstructured inputs such as news and market signals to generate actionable recommendations for inventory and procurement planning.

This approach enables SMEs to transition from reactive operations to proactive strategy, improving cost control, reducing inventory risks, and enhancing operational efficiency. By embedding explainable AI into everyday business workflows, StockWise serves as a proof-of-concept for how large language models can move beyond informational outputs to deliver tangible, measurable economic value in real-world SME environments.

3. Product Purpose

3.1. Main Goal of the System

The primary goal of StockWise is to fundamentally transform the inventory management paradigm for Malaysian SME traders from reactive monitoring to predictive intelligence. Rather than waiting for threshold alerts to trigger after stock levels have already become critical, StockWise proactively analyzes a comprehensive set of operational and market data to generate forward-looking recommendations that help businesses stay ahead of supply chain risks and capitalize on favorable procurement windows.

This transformation is achieved by combining traditional structured data analysis (inventory levels, sales velocity calculations, exchange rate tracking) with advanced AI-powered reasoning that incorporates unstructured market intelligence. The result is a system that not only tells users what their current inventory status is, but also explains what is likely to happen, why it is likely to happen, and what specific actions they should consider taking — complete with financial impact estimates, priority rankings, and multi-scenario strategy options. StockWise effectively serves as an AI-powered supply chain advisor that augments human decision-making rather than attempting to replace it.

3.2. Intended Users (Target Audience)

StockWise is designed to serve four primary user personas within Malaysian SME trading and import businesses:

1. **Warehouse Managers**: Responsible for day-to-day stock monitoring, warehouse capacity planning, and ensuring product availability. They benefit from AI-powered early warnings about potential stockouts and supply disruptions, enabling them to plan warehouse space allocation more effectively.
2. **Procurement Officers**: Tasked with sourcing products from international suppliers at optimal costs and timing. They benefit from exchange rate-aware procurement recommendations, supplier risk assessments, and reasoning chains that justify when and how much to order from each supplier.
3. **Business Owners**: Need a high-level strategic overview of inventory health, financial exposure, and growth opportunities. They benefit from the executive dashboard KPIs,

strategy validation scenarios, and financial impact summaries that support data-driven decision-making.

4. **Supply Chain Analysts**: Professionals focused on optimizing the end-to-end supply chain. They benefit from the comprehensive data integration, multi-factor trade-off analysis, and the ability to evaluate multiple procurement strategies simultaneously.

4. System Functionalities

4.1. Description

StockWise is implemented as a full-stack Next.js web application backed by a SQLite database for persistent data storage. The application follows a single-page application (SPA) architecture where all eight functional modules are accessible through a unified dashboard interface, with client-side state management controlling which module is displayed to the user. This architecture provides a seamless, app-like experience without page reloads, which is particularly important for maintaining context during AI-powered analysis workflows.

The technology stack comprises Next.js (React framework) for the frontend and API layer, SQLite with the better-sqlite3 library for data persistence, Recharts for data visualization, Tailwind CSS for responsive UI styling, and Z AI's GLM API for all AI-powered reasoning and recommendation generation. The server-side API routes handle data collection, prompt construction, GLM API communication, and response parsing, ensuring that sensitive API keys and complex data aggregation logic remain securely on the server.

4.2. Key Functionalities

The StockWise platform comprises eight integrated functional modules:

1. **Dashboard Overview**

   The dashboard serves as the command center of the StockWise platform, presenting four key performance indicator (KPI) cards that provide an at-a-glance summary of the business's inventory health. These KPIs include Total Inventory Value (the aggregate MYR value of all stock on hand), At-Risk Stock count (number of items below optimal reorder thresholds), AI Active Suggestions (count of pending AI-generated recommendations), and Estimated Savings (cumulative financial impact of AI-recommended actions). The dashboard provides immediate visibility into critical metrics, enabling quick assessment of overall supply chain status.

2. **Inventory Management**

   This module provides comprehensive stock level management with real-time visibility into current inventory quantities, reorder points, and calculated days-of-cover for each product. Users can manually adjust stock levels to reflect incoming shipments, returns, damages, or cycle count corrections. Each inventory item displays its current status (optimal, low, critical) based on automatically calculated thresholds derived from historical sales velocity data. The module also supports filtering by status and category, enabling warehouse managers to quickly identify items requiring immediate attention.

3. **Product & Supplier Management**

   This module offers full CRUD (Create, Read, Update, Delete) operations for both products and suppliers, serving as the master data management layer of the system. Each product record includes name, SKU, category, unit price, supplier reference, current stock level, and reorder point. Each supplier record tracks the supplier name, country of origin, and primary trading currency (USD, EUR, or CNY). The currency tracking feature is essential for the AI recommendation engine, as it enables the system to calculate procurement costs in MYR and factor exchange rate exposure into its analysis.

4. **Sales Velocity**

   The sales velocity module provides revenue visualization and trend analysis powered by Recharts. It displays interactive revenue charts showing sales trends over customizable time periods, calculated daily sales averages for each product, and the ability to log individual sales transactions with date, quantity, and revenue data. This module serves as the primary data source for the AI's demand forecasting calculations, as accurate sales velocity data is essential for generating reliable inventory recommendations. The visual charts help users quickly identify seasonal patterns, growth trends, and products with declining demand.

5. **Exchange Rate Management**

   This module tracks exchange rates for the three most relevant currencies for Malaysian importers: USD/MYR, EUR/MYR, and CNY/MYR. Users can manually update rates as

they change, and the system records the timestamp of each update. The exchange rate data is critical for the AI recommendation engine, as it enables the system to identify favorable or unfavorable procurement windows based on currency movements. A rate trend indicator shows whether each currency has appreciated or depreciated since the last update, helping procurement officers time their orders strategically.

6. **Market News Intelligence**

   The market news module provides a curated feed of relevant supply chain and trade-related news articles sourced from reputable financial and trade publications. Each news item includes a headline, summary, source attribution, and publication timestamp. Critically, this module serves as the unstructured data pipeline for the AI recommendation engine — when AI recommendations are generated, the most recent news articles are included as contextual input to the GLM model, enabling the AI to cross-reference current market events with inventory data. This integration of structured and unstructured data is the key differentiator that elevates StockWise above conventional inventory management tools.

7. **AI Recommendations**

   This is the core AI-powered module of StockWise. Users can trigger the generation of new AI recommendations by clicking the 'Generate New Insights' button, which initiates a comprehensive analysis pipeline. The system collects all relevant data (inventory levels, sales velocity, exchange rates, and market news), constructs a detailed prompt, and submits it to Z AI's GLM for multi-step reasoning analysis. The resulting recommendations include the product name, a specific recommended action, a priority level (high, medium, low), estimated financial impact, a detailed reasoning chain explaining the AI's analysis, and specific news article references that informed the recommendation. Each recommendation can be expanded to reveal its full reasoning chain, providing transparency into the AI's decision-making process.

8. **Strategy Validation**

   The strategy validation module enables users to perform AI-powered trade-off analysis by evaluating procurement strategies across three distinct scenarios: Conservative

(minimizing risk, prioritizing cash flow preservation), Balanced (moderate risk-taking with attention to both cost savings and supply security), and Aggressive (maximizing procurement savings by capitalizing on favorable conditions). For each scenario, the AI provides specific action items, expected financial outcomes, risk assessments, and a comparative analysis that helps business owners select the strategy that best aligns with their current financial position and risk appetite. The module effectively serves as a decision-support tool for capital allocation and procurement planning.

4.3. AI Model & Prompt Design

4.3.1. Model Selection

StockWise integrates with Z AI's GLM model via the provided ILMU API to serve as the system's core AI reasoning engine. The underlying GLM model was selected based on three key capabilities that are essential for the platform's functionality. First, GLM demonstrates strong multi-step reasoning capabilities, enabling it to perform the complex analytical chains required to cross-reference inventory data, sales trends, exchange rates, and market news simultaneously. Second, GLM reliably generates well-structured JSON output, which is critical for the system's ability to parse AI responses into actionable recommendation objects with specific fields. Third, GLM excels at integrating and reasoning across both structured quantitative data (numbers, percentages, dates) and unstructured qualitative data (news articles, market commentary), which is the foundational capability that makes StockWise's news-informed recommendations possible.

4.3.2. Prompting Strategy

StockWise employs a multi-step agentic prompting strategy that maximizes the quality and relevance of AI-generated recommendations. The system first collects all relevant data server-side — inventory levels, recent sales transactions, current exchange rates, and the latest market news articles — before constructing a comprehensive, structured prompt that instructs the GLM to perform holistic analysis. The prompt is designed to elicit

specific response formats, including JSON-structured recommendations with reasoning chains, news references, priority classifications, and financial impact estimates. This structured prompting approach ensures consistency in AI output format while allowing flexibility in the analytical reasoning process.

4.3.3. Context & Input Handling

The system collects all relevant data server-side before sending it to the GLM model, ensuring that no sensitive API keys or complex data processing logic is exposed on the client side. Market news (up to 6 recent articles) is embedded as contextual input within the prompt, providing the AI with the unstructured data necessary for news-informed analysis. The typical prompt size is approximately 2000 tokens, which includes the system instructions, structured data payload, and news context. To handle cases where data volume might exceed the token limit, the system implements an intelligent truncation strategy: if the total data payload exceeds the limit, only the most recent 30-day sales data is included, ensuring that the most temporally relevant information is always prioritized.

4.3.4. Fallback & Failure Behavior

The system implements several layers of error handling to ensure robustness. If the AI returns invalid JSON that cannot be parsed, the system returns an HTTP 500 error response that includes the raw AI response for debugging purposes, enabling developers to diagnose prompt or response format issues. If the AI returns partially valid content — for example, some well-formed recommendations mixed with off-topic or malformed content — the system attempts to salvage and save whatever valid recommendations it can parse, ensuring that users still receive useful insights even when the AI response is not perfect. Rate limiting is applied at the API level to prevent abuse and manage API costs, protecting against both inadvertent excessive usage and potential denial-of-service scenarios.

5. User Stories & Use Cases

5.1 User Stories

5.1.1. User Story 1

"As a warehouse manager, I want AI to analyze market news alongside my inventory data so that I receive proactive recommendations before stock runs low."

This user story captures the core value proposition of StockWise. The warehouse manager is the primary operational user who needs to maintain optimal stock levels across multiple product categories. By integrating market news analysis into the recommendation pipeline, StockWise enables this user to receive warnings about potential supply disruptions — such as shipping delays due to port congestion or raw material shortages affecting a supplier — before they manifest as actual stockout events. This proactive capability transforms the warehouse manager's role from reactive firefighter to strategic planner, enabling better space utilization and more efficient labor scheduling.

5.1.2. User Story 2

"As a procurement officer, I want to see AI reasoning chains so that I understand why a recommendation was made and can make informed decisions."

Trust in AI-generated recommendations is essential for user adoption, particularly in procurement decisions that involve significant financial commitments. This user story addresses the 'black box' problem that plagues many AI systems by requiring every recommendation to include a detailed reasoning chain that explains the specific data points, trends, and market conditions that led to the recommendation. The procurement officer can review this reasoning, validate the AI's logic against their own domain expertise, and make a more confident and informed decision. This transparency also

enables the officer to identify potential blind spots in the AI's analysis, creating a collaborative human-AI decision-making workflow.

5.1.3. User Story 3

"As a business owner, I want to run strategy simulations that consider multiple factors (cash flow, exchange rates, supply chain risks) so that I can prioritize actions within budget constraints."

Business owners need to make strategic decisions that balance multiple competing priorities within finite budget constraints. This user story drives the Strategy Validation module, which allows the business owner to evaluate different procurement strategies across conservative, balanced, and aggressive scenarios. Each scenario analysis considers the interplay between available cash flow, current exchange rate conditions, supply chain risk factors, and demand forecasts, enabling the owner to select the strategy that best aligns with their risk tolerance and financial objectives. The multi-factor analysis ensures that no single variable dominates the decision, providing a holistic view of the trade-offs involved in each strategic option.

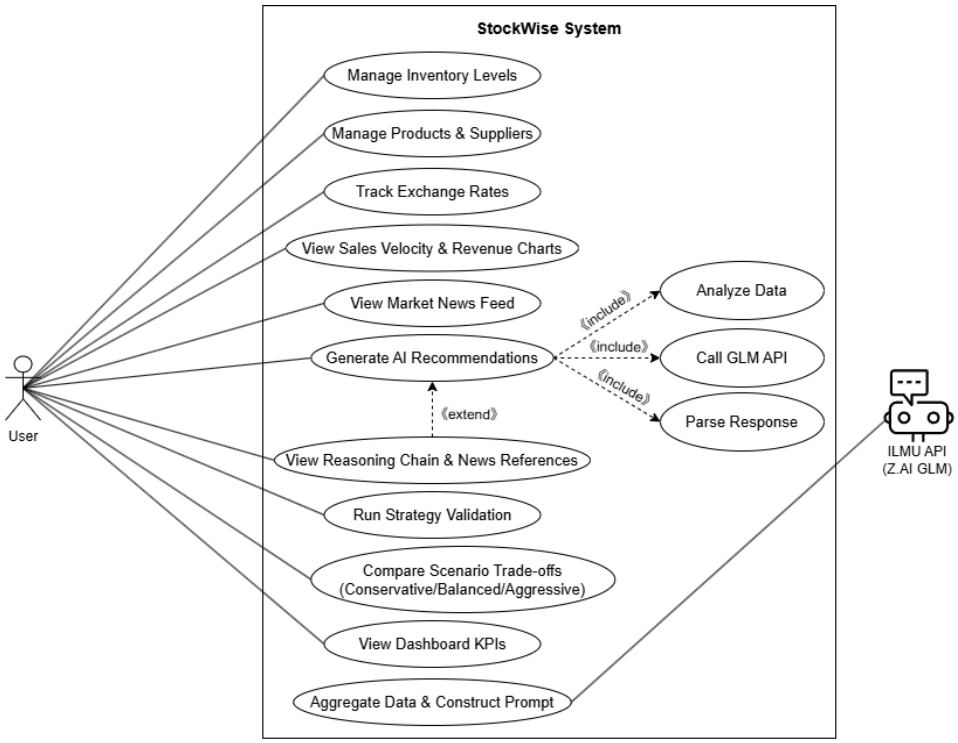
5.2 Use Cases (Main Interactions)

The primary use case for StockWise's AI capability is the **Generate AI Recommendations flow**, which represents the end-to-end process from user initiation to actionable insight delivery. The complete workflow proceeds through the following steps:

- The user navigates to the AI Recommendations module and clicks the 'Generate New Insights' button to initiate a new analysis cycle.
- The backend API route activates and begins collecting all relevant data: current inventory levels for all products, recent sales transaction data (up to 30 days), the latest exchange rate entries for USD/MYR, EUR/MYR, and CNY/MYR, and the most recent market news articles (up to 6 articles).

- The system constructs a comprehensive prompt that includes system instructions, the collected structured data payload formatted for AI comprehension, and the embedded market news articles as unstructured context.
- The constructed prompt is submitted to Z.AI's GLM API, where the model performs multi-step reasoning analysis across all data dimensions simultaneously.
- The GLM returns a JSON-structured response containing individual recommendation objects, each with specific fields including product reference, recommended action, priority level, estimated financial impact, reasoning chain, and news references.
- The backend parses the JSON response, validates each recommendation object, and saves valid recommendations to the SQLite database with their associated reasoning chains and news references intact.
- The frontend receives the saved recommendations and displays them in an interactive card-based layout. Each recommendation card shows the key details at a glance and can be expanded to reveal the full reasoning chain and linked news articles, providing complete transparency into the AI's analytical process.

5.3 Use Case Diagram



6. Features Included (Scope Definition)

StockWise includes the following features in its minimum viable product (MVP) release. Each feature has been designed and implemented to deliver maximum value to the target users while maintaining a cohesive and intuitive user experience:

- **AI-Powered Inventory Recommendations with News-Informed Reasoning:** AI-powered inventory recommendations that leverage Z AI's GLM to perform multi-step reasoning analysis across inventory data, sales trends, exchange rates, and market news, producing actionable insights with full transparency through included reasoning chains.
- **Multi-Factor Trade-Off Analysis with Scenario Comparison:** A dedicated strategy validation module that evaluates procurement scenarios across conservative, balanced, and aggressive profiles, considering cash flow constraints, exchange rate exposure, supply chain risks, and opportunity costs to help users select the optimal approach.
- **Real-Time Dashboard with Interactive KPI Cards:** An executive dashboard featuring four interactive KPI cards (Total Inventory Value, At-Risk Stock Count, AI Active Suggestions, and Estimated Savings) that provide an at-a-glance summary of overall inventory health and AI-driven financial impact.
- **Multi-Currency Exchange Rate Tracking:** Support for tracking exchange rates across USD/MYR, EUR/MYR, and CNY/MYR with manual update capability, timestamp recording, and trend indicators that feed directly into the AI recommendation engine for procurement timing optimization.
- **Market News Intelligence Feed:** A curated market news feed covering supply chain disruptions, trade policy changes, and commodity price trends. News articles are integrated into the AI analysis pipeline as unstructured context, enabling news-informed inventory recommendations.
- **Sales Velocity Tracking with Revenue Visualization:** Comprehensive sales tracking with revenue visualization charts powered by Recharts, daily sales average calculations, and individual sales transaction logging. Sales velocity data serves as the foundation for AI demand forecasting and days-of-cover calculations.
- **Product and Supplier CRUD Management:** Full CRUD operations for managing product catalogs (name, SKU, category, unit price, supplier) and supplier profiles (name, country, currency), serving as the master data layer that feeds all AI analysis and reporting modules.
- **Inventory Adjustment with Status Tracking:** Manual stock adjustment capabilities with automatic status classification (optimal, low, critical) based on calculated thresholds, enabling warehouse managers to maintain accurate inventory records and quickly identify items requiring attention.

7. Features Not Included (Scope Control)

The following features have been deliberately excluded from the MVP scope to maintain focus on core value delivery and achievable development timelines within the hackathon context. These represent potential future enhancements for post-hackathon development:

- **Real-Time News Scraping:** The MVP uses a curated static news feed rather than implementing real-time web scraping. This decision prioritizes reliability and content quality over freshness. A production version would integrate with news APIs or implement scheduled scraping to maintain a continuously updated feed.
- **Multi-User Authentication and Role-Based Access:** StockWise does not include user authentication, login systems, or role-based access control in the MVP. The application is designed as a single-user tool. Multi-user support with different permission levels (admin, manager, viewer) is planned for future iterations.
- **Automated Purchase Order Generation:** While the AI recommends procurement actions, the MVP does not automatically generate or send purchase orders to suppliers. Users must manually act on recommendations. Automated PO generation with supplier communication integration is a natural extension for production deployment.
- **Mobile Native Applications:** StockWise is a responsive web application designed for desktop and tablet browsers. Native mobile applications for iOS and Android are not included in the MVP scope but represent a high-priority future development pathway given the mobile-first preferences of many SME operators.
- **External ERP/Accounting System Integration:** The MVP operates as a standalone system with no integration points to external enterprise resource planning (ERP), accounting, or point-of-sale (POS) systems. Data import/export functionality and API-based integration with tools like Xero, QuickBooks, or SAP Business One would be required for enterprise deployment scenarios.

8. Assumptions & Constraints

8.1 LLM Cost Constraint

Each AI recommendation generation call consumes approximately 2000 tokens, encompassing the system prompt, structured data payload, news context, and the generated response. Rate limiting is applied at the application level to restrict usage to a maximum of 10 API calls per hour, which balances the need for timely insights with API cost management and prevents excessive usage during testing or demonstration scenarios. No response caching mechanism has been implemented in the MVP, meaning that each generation request triggers a fresh API call even if the underlying data has not changed. Implementing a cache with data-change-based invalidation would be a high-value optimization for future iterations, potentially reducing API costs by 60–80% for users who generate recommendations frequently.

8.2 Technical Constraints

The SQLite database was selected for the MVP due to its zero-configuration deployment model, which eliminates the need for a separate database server and simplifies the development and deployment process. However, SQLite is not suitable for high-concurrency enterprise environments or deployments requiring multi-server horizontal scaling. For production deployment with multiple simultaneous users, migration to PostgreSQL or MySQL would be necessary. Additionally, the single-page application architecture (all modules accessible through one route with client-side state management) simplifies navigation but limits support for deep linking and browser back-button navigation to module states.

8.3 Performance Constraints

AI recommendation generation typically requires 5 to 15 seconds to complete, depending on the volume of data being analyzed and current API response times. During this processing period, the user interface displays a loading state but does not provide real-time progress updates. No WebSocket connection is maintained between the client and server, meaning that the user must remain on the AI Recommendations page and wait

for the response to render. For the MVP, this synchronous request-response pattern is acceptable, but a production system would benefit from asynchronous job processing with WebSocket notifications to allow users to continue working while AI analysis completes in the background.

8.4 User Input

All data entry in the MVP is performed manually through web form interfaces. There is no barcode scanning integration, no point-of-sale (POS) system integration, and no bulk data import functionality (CSV/Excel upload). While manual entry ensures data accuracy for demonstration purposes and eliminates external system dependencies, it creates a practical limitation for businesses with large product catalogs or high transaction volumes. A production system would require barcode scanning integration for inventory counts, POS integration for automated sales recording, and bulk import/export capabilities for initial data population and periodic synchronization with external systems.

9. Risks & Questions Throughout Development

9.1 Recommendation Homogeneity (The "Generic Advice" Risk)

- **Risk**: How are we ensuring that Z.AI's GLM does not generate generic, one-size-fits-all recommendations (e.g., simply saying "reorder when low") instead of providing true, context-aware trade-off analysis based on the specific interplay of FX rates and news for a given SKU?
- **Mitigation**: The system employs strict, highly-structured prompt engineering that forces the GLM to explicitly calculate and reference specific data points (e.g., exact MYR depreciation percentages, specific news headlines) in its reasoning chain before outputting a recommendation. If the GLM fails to cite specific data, the UI flags the recommendation as "Low Confidence."

9.2 Data State Consistency (The "Moving Target" Risk)

- **Risk:** Because the Z.AI API takes 5 to 15 seconds to process a prompt, what happens if a user updates an inventory level or exchange rate while the AI is currently analyzing the old data? The resulting recommendation would be based on a stale database state.
- **Mitigation:** The backend implements a "Snapshot Architecture." The moment the user clicks 'Generate New Insights', the system queries the SQLite database and freezes that exact data state in memory. The Z.AI prompt is constructed entirely from this frozen snapshot. Any user changes made during the 15-second API wait time are ignored for that specific generation cycle, ensuring logical consistency between the AI's advice and the data it claims to be based on.

9.3 Unstructured News Misinterpretation (The "False Signal" Risk)

- **Risk:** What happens if Z.AI misinterprets a nuanced or unrelated market news article (e.g., a macro-economic forecast) and falsely correlates it to a specific SKU, triggering a panic-buy recommendation that ties up the SME's cash flow unnecessarily?
- **Mitigation:** StockWise mandates Explainable AI (XAI) principles. Every recommendation must include the exact news headline it used. Furthermore, the system relies on a curated, domain-specific news feed rather than raw web scraping, reducing the chance of irrelevant data poisoning the AI's logic. The UI prominently displays a disclaimer requiring human validation of the AI's cited sources before executing procurement.